WK1-PM1 – Cybersecurity Lab Setup Report

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Program	Cybersecurity Internship, Networkwalks
Task	WK1-PM1 – Lab Setup (Week 1, Project Module 1)
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Objective

This task sets up a cybersecurity practice lab where Kali Linux acts as the attacker machine. Kali must live on a NAT network in 10.0.0.0/24 with the fixed address 10.0.0.2/24, reach the Internet, and support clipboard sharing, drag-and-drop and a shared Downloads folder from the host.

How the lab was built: the setup was done together with my lab partner, Selvi Melani, on her windows laptop running VMware Workstation. We configured each step jointly, so the screenshots in this report come from that shared session.

Lab environment

We used VMware Workstation as the hypervisor rather than VirtualBox and matched every VirtualBox requirement with a VMware setting.

Component	Value
Host machine	Windows laptop (lab partner's)
Hypervisor	VMware Workstation (version: *fill in*)
Attacker VM	Kali Linux, ISO install (version: *fill in*)
VM resources	6 GB RAM · 4 vCPU · 40 GB disk
Lab network	VMnet8 (NAT) · 10.0.0.0/24
Gateway / DNS	10.0.0.1 / 8.8.8.8
DHCP pool	10.0.0.128 – 10.0.0.254
Kali IP	10.0.0.2/24 (static)

Task wording (VirtualBox)	What we used (VMware Workstation)
NATNetwork in 10.0.0.0/24	VMnet8 (NAT), subnet 10.0.0.0/24
Adapter on NATNetwork	Network Adapter → Custom: VMnet8 (NAT)
Clipboard & drag/drop	Options → Guest Isolation
Shared folder /downloads	Options → Shared Folders → /mnt/hgfs/Downloads in Kali

Requirements compliance checklist

#	Requirement from WK1-PM1	Result	See
1	VirtualBox as base	Met using VMware Workstation equivalent	Fig. 1, 5
2	Kali Linux as attacking machine	Met	Fig. 1, 6, 7
3	Subnet 10.0.0.0/24	Met	Fig. 2
4	NAT network on 10.0.0.0/24	Met (VMnet8)	Fig. 2 – 5
5	Clipboard & drag/drop enabled	Met	Step 7
6	Host Downloads shared	Met	Step 8
7	Kali IP 10.0.0.2/24	Met	Fig. 6, 7
8	Full Internet access	Met	Fig. 7

Configuration walkthrough

1. Kali VM created in VMware

We created the Kali VM from the ISO and gave it 6 GB of RAM, 4 processors and 40 GB of disk. At first the adapter was on VMware's default NAT.

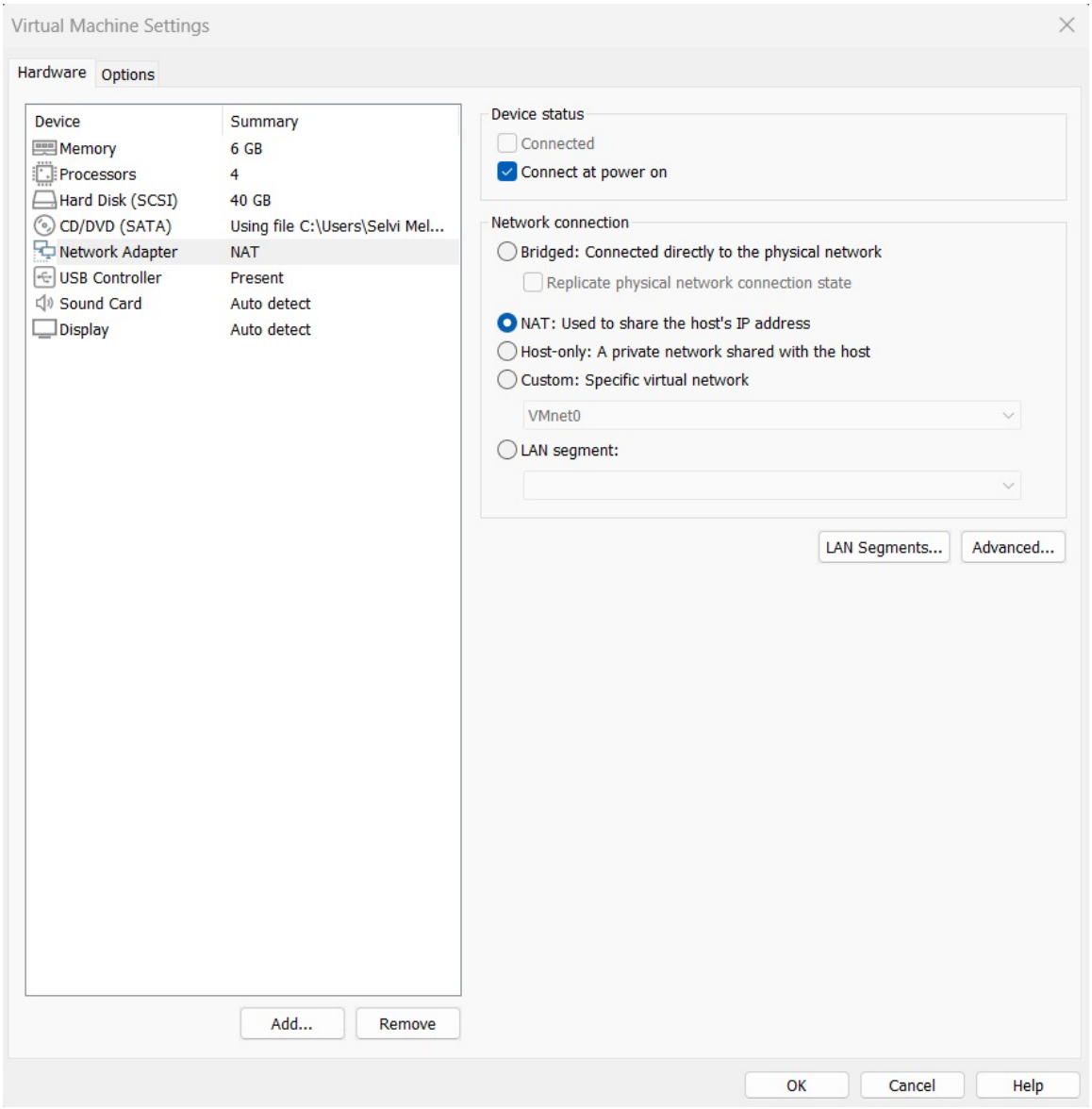


Figure 1 – Kali VM hardware, adapter on default NAT

2. NAT network moved to 10.0.0.0/24

Using the Virtual Network Editor, we changed VMnet8 (NAT) to subnet 10.0.0.0, mask 255.255.255.0. VMnet8 is VMware's counterpart of a VirtualBox NAT Network.

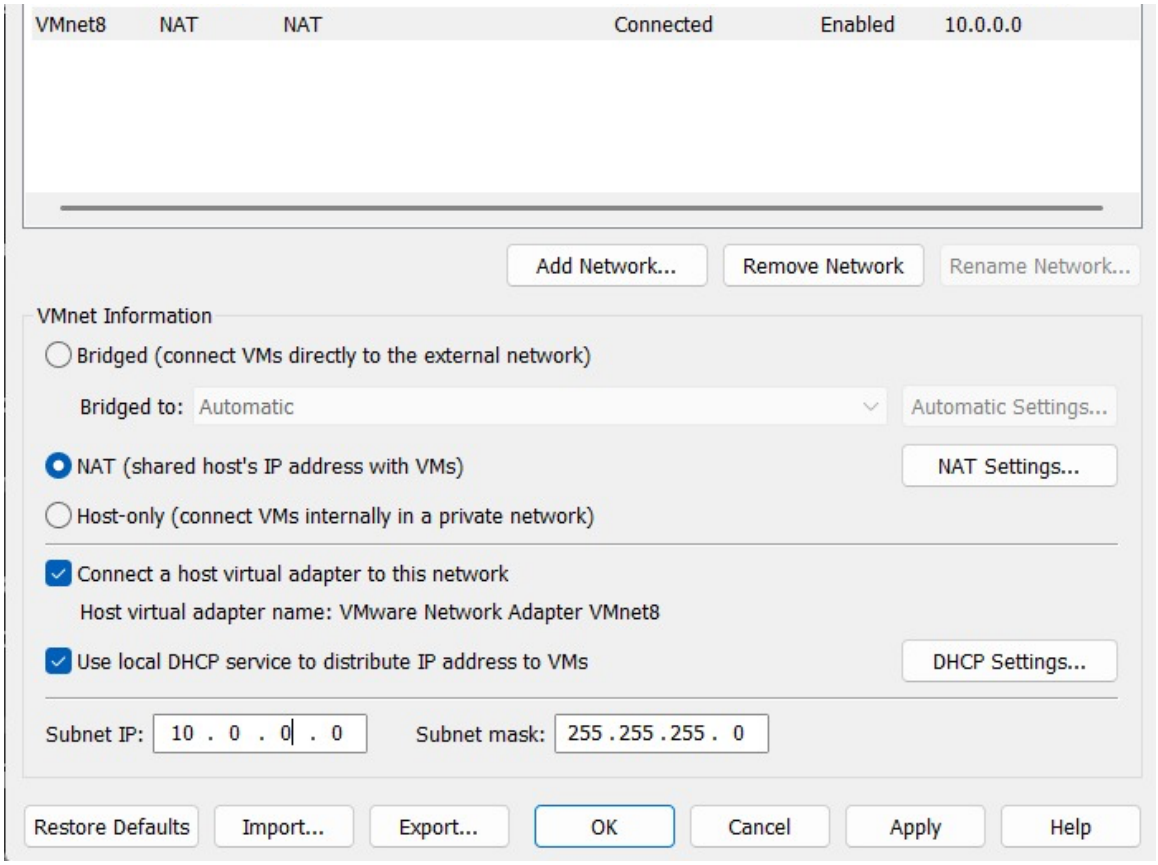


Figure 2 – Virtual Network Editor, VMnet8 on 10.0.0.0/24

3. DHCP pool checked

VMnet8's DHCP only leases 10.0.0.128 to 10.0.0.254. That leaves the lower range, including 10.0.0.2, for the static addresses used in the lab design.

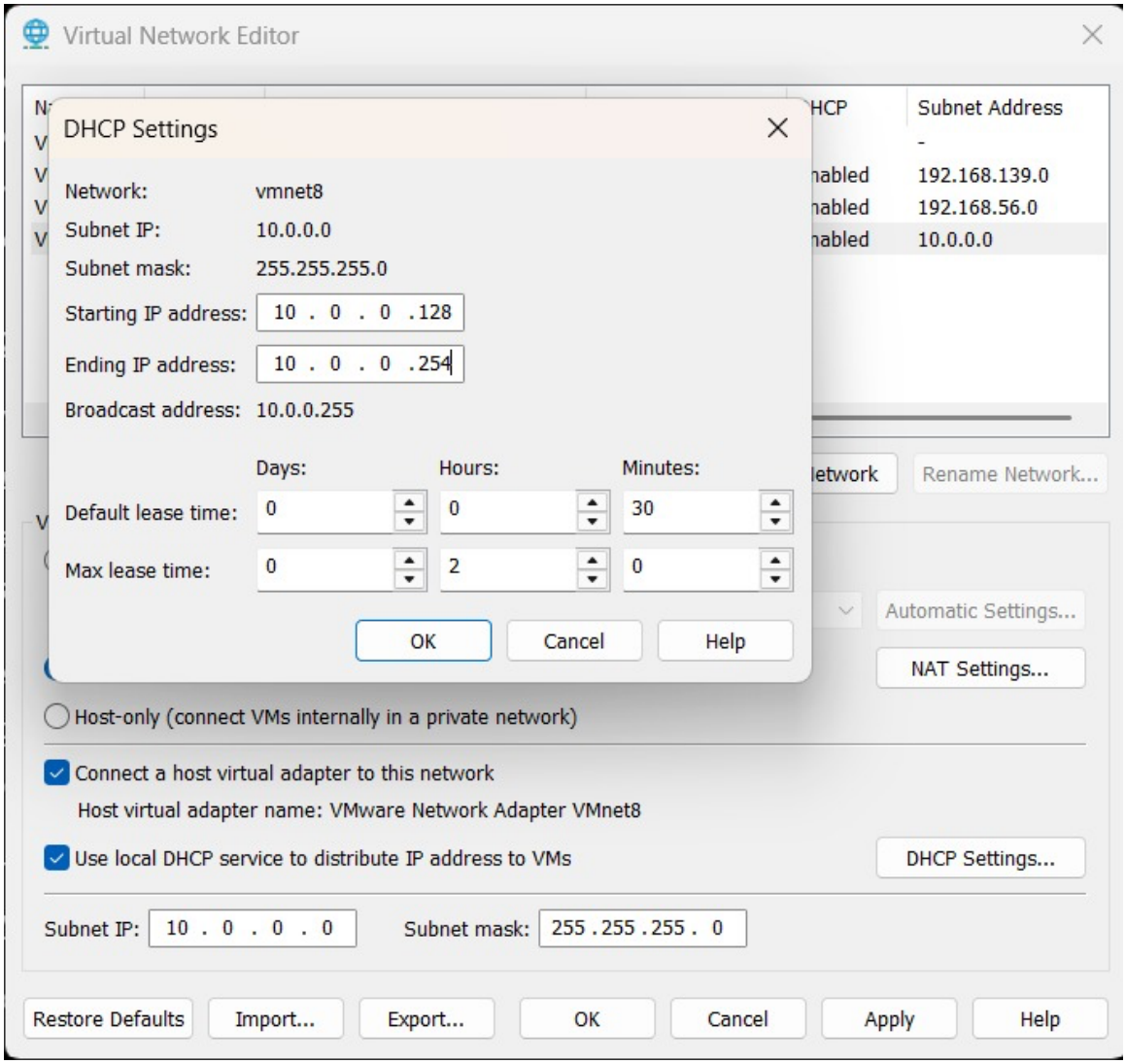


Figure 3 – DHCP settings for VMnet8

4. Gateway set to 10.0.0.1

Under NAT Settings we set the gateway to 10.0.0.1, which every VM in the lab uses as its default gateway.

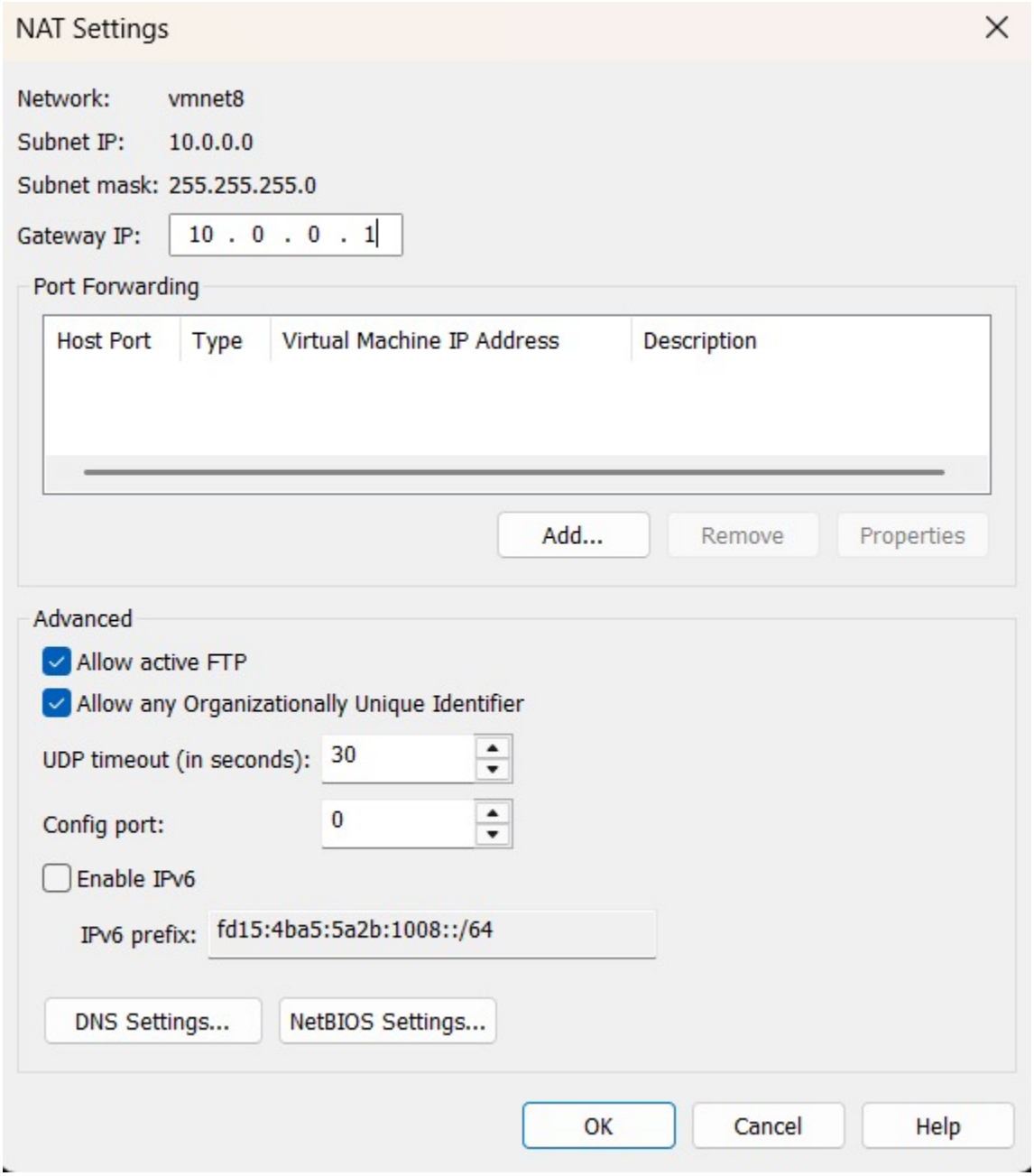


Figure 4 – NAT Settings, gateway 10.0.0.1

5. Kali adapter pointed to VMnet8

We changed the Kali adapter from plain NAT to Custom → VMnet8 (NAT) so it is bound to the lab network directly.

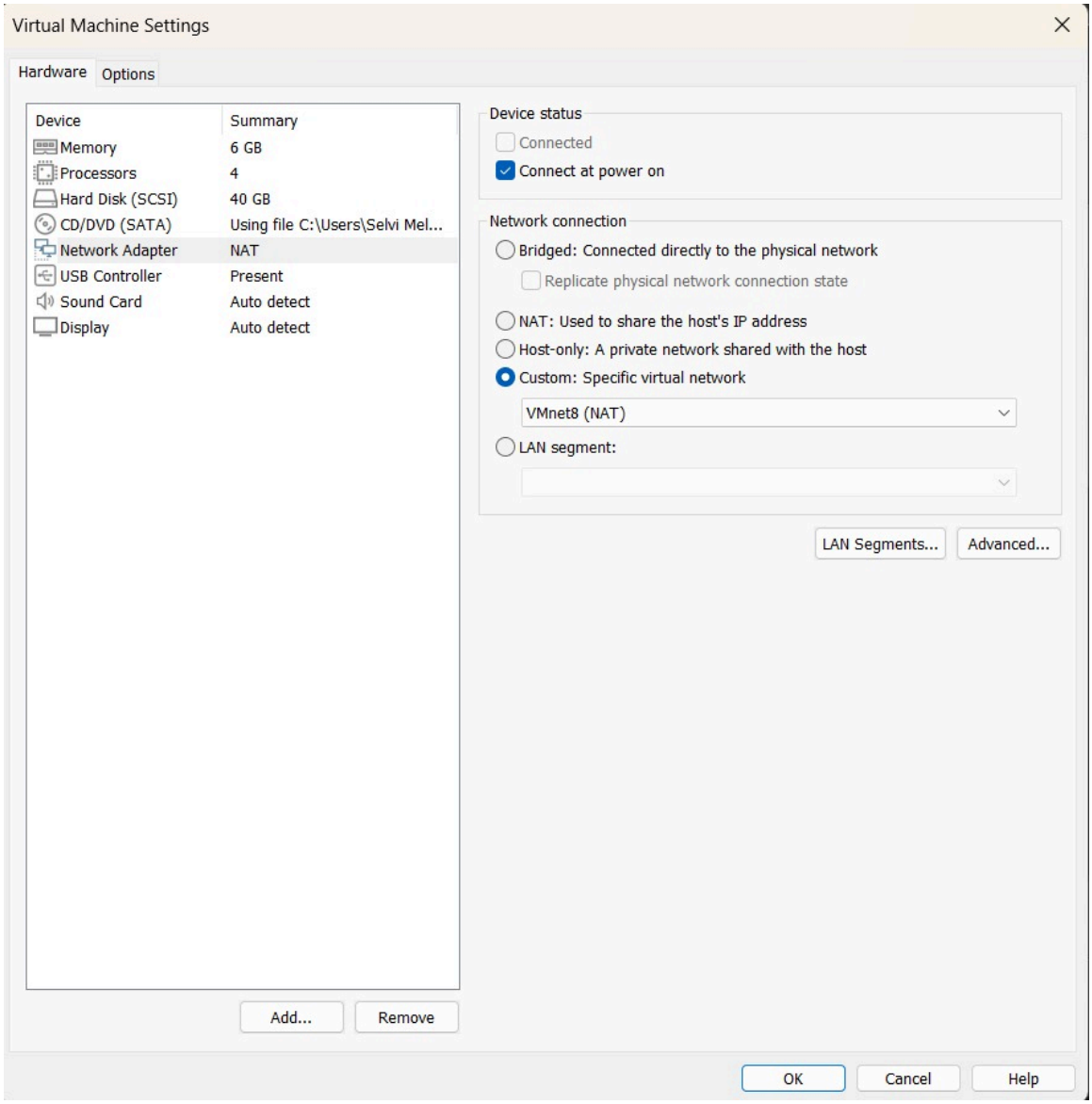


Figure 5 – Kali adapter on Custom: VMnet8 (NAT)

6. Static IP 10.0.0.2/24 in Kali

We opened `nmtui`, edited "Wired connection 1" and set IPv4 to Manual: 10.0.0.2/24, gateway 10.0.0.1, DNS 8.8.8.8.

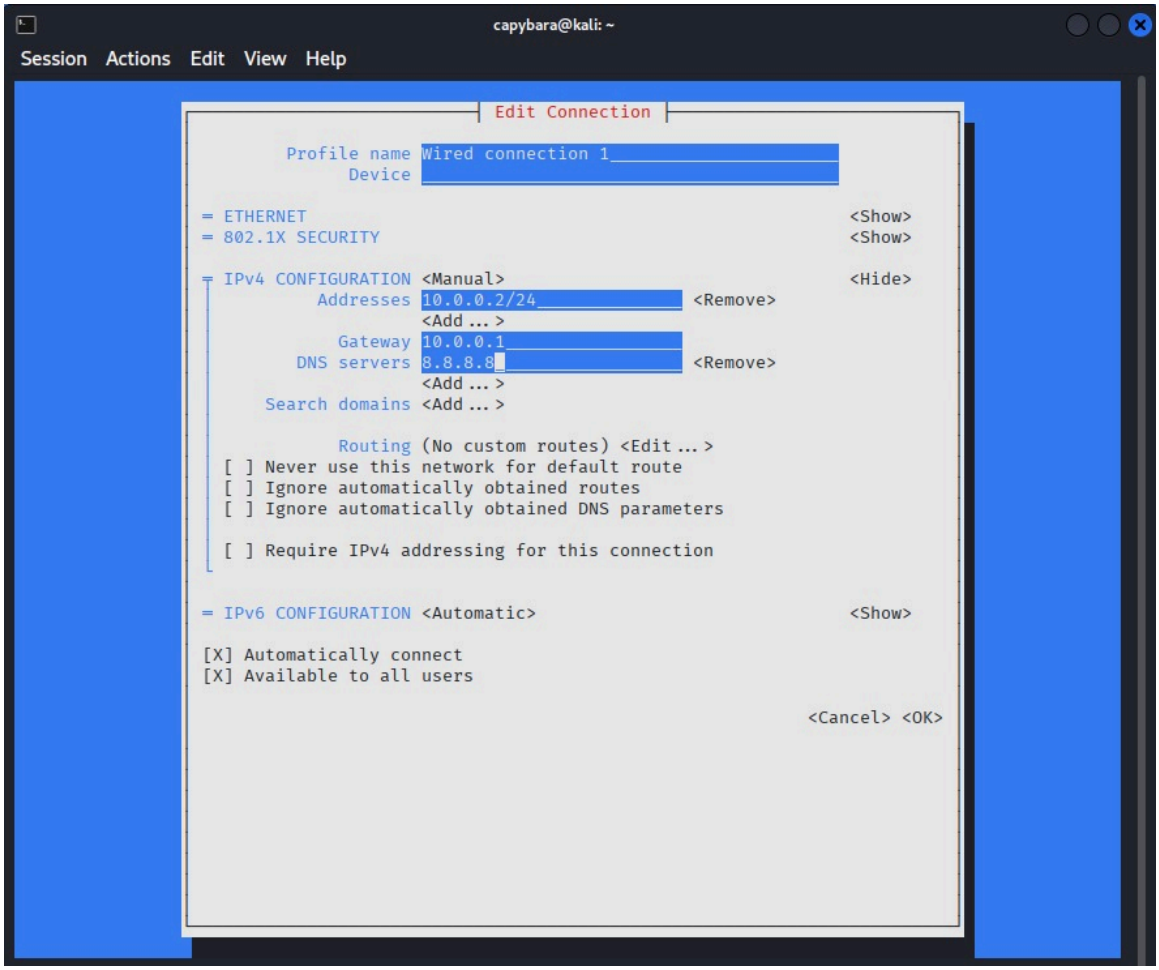


Figure 6 – nmtui manual IPv4 settings

After `sudo nmcli connection up "Wired connection 1"`, `ip -br addr` showed eth0 UP at 10.0.0.2/24 and `ip route` showed the default route through 10.0.0.1.

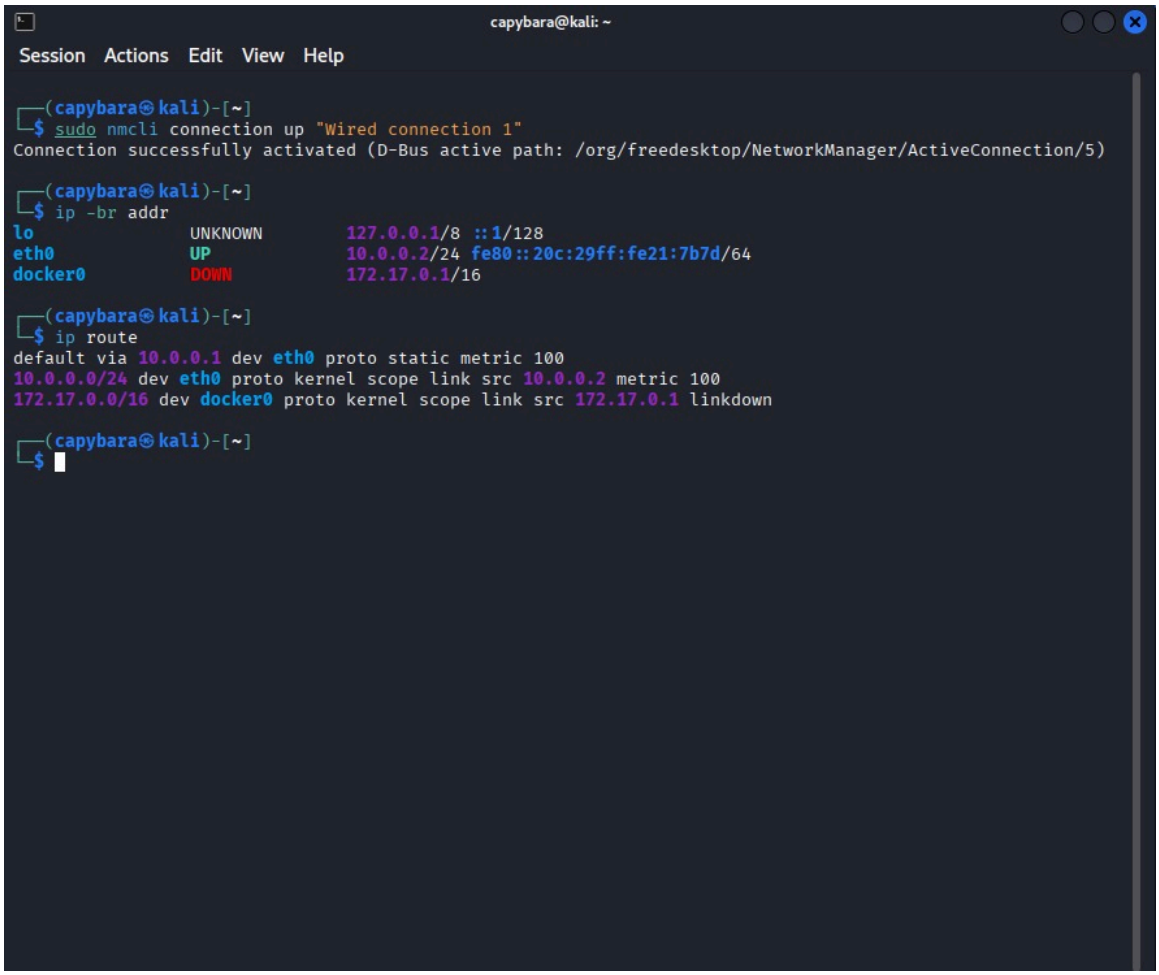


Figure 7 – ip addr and ip route output in Kali

7. Clipboard and drag-and-drop

We shut down Kali, opened VM Settings → Options → Guest Isolation and ticked both "Enable drag and drop" and "Enable copy and paste". Once Kali was back up, copy-paste worked between the host and the Kali terminal in both directions, and dragging a file from the host desktop copied it into the VM.

8. Shared Downloads folder

In VM Settings → Options → Shared Folders we chose "Always enabled" and added the host's Downloads folder. Kali mounts VMware shares under /mnt/hgfs, and `ls /mnt/hgfs/Downloads` showed the files from the host.

Testing & results

We ran these checks in Kali to confirm addressing, routing, Internet access, the shared folder and clipboard integration.

Check	Command / action	Result
Connection	`sudo nmcli connection up "Wired connection 1"`	Activated successfully (Fig. 7)
Address	`ip -br addr`	eth0 UP at 10.0.0.2/24 (Fig. 7)
Route	`ip route`	default via 10.0.0.1 (Fig. 7)
Gateway	`ping -c 4 10.0.0.1`	Replies received
Internet	`ping -c 4 8.8.8.8`	Replies received
DNS	`ping -c 4 google.com`	Hostname resolved, replies received
Shared folder	`ls /mnt/hgfs/Downloads`	Host files listed
Clipboard	Copy-paste host ↔ Kali	Works both ways
Drag-and-drop	File from host into Kali	File copied into the VM

Every check passed, which confirms the lab meets all eight requirements.

Observations & lessons learned

- **Hypervisor differences.** VirtualBox and VMware name the same ideas differently: a VirtualBox NAT Network is a VMware VMnet with NAT, and Shared Clipboard is Guest Isolation. Mapping them helped me understand what each setting actually does instead of just following screenshots.
- **Why 10.0.0.1 as gateway.** VMware's default puts the NAT gateway on .2. Since the task reserves 10.0.0.2 for Kali, the gateway had to move to 10.0.0.1, otherwise Kali and the gateway would clash.
- **Static vs DHCP range.** Keeping static lab addresses (10.0.0.2 – 10.0.0.99) below the DHCP pool (.128 – .254) prevents conflicts when Windows and Android VMs join in Phase 2.
- **If Internet fails.** The course's fix for Kali 2026.x is to set `ipv4.dad-timeout 0` on "Wired connection 1" and then restart the connection with nmcli.

Conclusion

The lab meets the WK1-PM1 specification: Kali Linux runs as the attacker machine at 10.0.0.2/24 on a NAT network (10.0.0.0/24) with gateway 10.0.0.1 and full Internet access, and clipboard, drag-and-drop and the shared Downloads folder all work. It is ready for Phase 2, where target machines will be added to the same subnet.